Shuvam Banerji Seal

Aspiring Computational Chemist | AI/ML Researcher | DeepTech & AI-Fintech Co-Founder

BS-MS Student (Chemistry Major, Computer Science Minor) – Indian Institute of Science Education and Research, Kolkata

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Publications

AgriIR: Scalable Framework for Domain-Specific Knowledge Retrieval *ECIR 2026*
S. Banerji Seal, A. Poddar, A. Mishra, D. Roy. Advanced IR framework for agricultural knowledge accessibility in developing regions with domain-optimized retrieval algorithms.
DOI: 10.1007/978-3-032-21324-2_37

Hierarchical Opinion Classification using LLMs *FIRE 2025*
S. Banerji Seal, A. Mishra, U. Ghosh. Parameter-efficient fine-tuning of Gemma with custom two-layer classification head and class-weighted cross-entropy loss for hierarchical opinion labels.
https://ceur-ws.org/Vol-4173/T10-3.pdf

Query Reformulation & Layered Retrieval for ToT *TREC 2024 Proceedings*
S. Banerji Seal, S. Adhikary, S. Sar, D. Roy. Four-step query reformulation with two-layer BM25 retrieval for known-item retrieval; Recall@1000: 0.8067.
https://trec.nist.gov/pubs/trec33/papers/IISER-K.tot.pdf

Computational Modeling of [VO(SALIEP)(DTP)] *2024–25 (to be submitted)*
S. Banerji Seal, guided by Dr. S. Roy. DFT methods (B3LYP) for MO energy calculations, electronic structure analysis, and water reduction catalyst performance prediction.

Research Experience

Hybrid RAG for Verifiable Answer Synthesis *July 2025*
Under Dr. D. Roy, IISER-K. End-to-end RAG pipeline for IISER-K intranet: automated Selenium scraping, OCR (PyMuPDF, EasyOCR), Qwen3-4B embeddings in FAISS, hybrid BM25 + dense retrieval via Reciprocal Rank Fusion, Streamlit UI with inline citations.

Advanced Retrieval for TREC Tip-of-Tongue *2024*
Under Dr. D. Roy, IISER-K. Multi-layer BM25 filtering in Lucene/Java with dynamic search-domain contraction; transformer-based query expansion via LLMs achieving results on par with modified DPR models.

Industrial Experience

Research Intern, HistoXai (Astroloop Technologies Pvt. Ltd.) *May–Jul 2025*
Bangalore. Comparative analysis of 30+ digital histopathology QA tools (HistoQC, PathProfiler, GrandQC, HistoROI, FASTPathology) evaluating architectures, datasets, and performance metrics.
Tech: Python, OpenCV, PyTorch, Scikit-learn, Pandas.

Tutorials Developed

Complete C Programming Course *2025*
Open-source 20-module curriculum: fundamentals to advanced topics including Network Programming, ML in C, and GUI development with GTK4.

Python Course for Beginners *2025*
Progressive learning resource: core concepts to local LLM deployment, with interactive notebooks, real-world projects, and database integration.

Shaders for Beginners *2026*
Comprehensive educational repository for graphics shader programming from first principles: 10+ progressive lessons covering OpenGL, GLSL, Vulkan intro, audio-reactive shaders, with C and C++ implementations.

Selected Projects

Solvation of C60 NanoParticles in Water *2025*
Computational science project: LAMMPS molecular dynamics simulations of C60 solvation with AIREBO potentials, automated feature extraction (RDF, coordination numbers), and ML prediction pipeline (GPR, NN, XGBoost) for system properties at novel parameter values.

LAMMPS Data Web Viewer *2025*
Web-based 3D visualization tool for LAMMPS data files using React, TypeScript, Three.js/R3F, and Flask backend. Features file upload, paste, theme switching, material customization, and CI/CD deployment to GitHub Pages.

Legal Document Retrieval RAG *Feb 2025*
LLaMA-3.2, Nomic embeddings, ChromaDB, Mistral-OCR for legal texts via Streamlit.

Agentic Database Builder *Aug 2024*
AI-driven autonomous DB construction: LLM-based schema generation, data validation, scalable curation.

WeLearn Bot in C *2025*
Multi-threaded CLI + GTK4 GUI with encrypted credentials and bulk downloads.

E-Commerce GUI in C *2024*
GTK4 + SQLite3 with BM25 chatbot. Under Dr. K. Ghosh, IISER-K.

Event Coupon System · Spherical Harmonics *2023–25*
Flask + OAuth 2.0 + QR verification; 3D quantum orbital visualizer (Flask + Plotly).

TCP Monitor · Wi-Fi Optimizer · Coord. Chem. Sim. *2024–25*
System utilities, network optimization, scientific simulations. SMC Canteen (Django, 2000+ users).

Explore more projects: GitHub/Shuvam-Banerji-Seal · Portfolio Website

Research Libraries & Frameworks

Fernholz SPT Library *Ongoing*
Python library for Stochastic Portfolio Theory: diversity-weighted portfolios, relative arbitrage using stochastic calculus and continuous semimartingales.

LAMMPS Data Web Viewer *2025*
Web-based 3D molecular dynamics visualization (React, Three.js, Flask) for interactive atom/bond rendering. Active web service.

Entrepreneurial Ventures

Co-Founder & Lead Developer, Synapse *2025–Present*
AI-fintech startup funded by **MeitY Startup Hub (GENESIS)**, incubated at RISE Foundation, IISER-K. Democratizing ML/AI tools for smart trading alerts across stocks, crypto, and mutual funds. DPIIT registration in progress.

Co-Founder & CTO, UnderWater AI *2025–Present*
Deeptech startup funded by **MeitY Startup Hub (GENESIS)**. Deep neural networks for underwater image enhancement and early fusion learning for marine species identification.

Achievements & Honors

Hackathons

• **Capital One Launchpad (Top 14 / 5,073 teams)** *2025*
Developed multi-modal RAG platform for Indian agriculture (Team Fibonacci), Capital One India.
– Built agentic AI advisor supporting 20+ Indian languages with scientifically cited outputs using RAG, LangChain, Gemma, DeepSeek.
– Engineered autonomous data pipeline creating a novel 15k+ document dataset, released on Hugging Face.

• **StatusCode1 – 1st Rank (GIAN Track)** *2024*
AI search engine for GIAN's Abandoned US Patents. Nomic Embeddings + similarity search for natural language patent queries.
– Created web-scraping pipeline using Selenium, BeautifulSoup, and fake_useragent.

• **StatusCode0 – 1st Rank (MATLAB Track)** *2023*
Domestic waste type data analysis tool for a proposed start-up solution, IIIT-Kalyani.

National Level Basic Sciences Competitions

• **ChemEnigma (1st Rank)** – 72-hour chemistry contest (theory, experimental, concept-presentation), IISc Bangalore (2025)
• **All Bengal Chemistry Quiz (2nd Runners Up)** – Presidency University (2025)
• **Mimansa (Zonal Topper)** – Mathematical problem solving, IISER Pune (2024)
• **NAEST (Zonal Runners Up)** – Experimental skill test using homely items, IIT Kanpur & Shiksha Sopan (2023)

Competitive Examinations

• Qualified **JEE Mains and Advanced** – Ranked in Top 0.1% of candidates (2022)
• Qualified **IAT (IISER Aptitude Test)** – Ranked in Top 0.06% of candidates (2022)
• Qualified **WBJEE** – Ranked in Top 0.05% of candidates (2022)

Scholarships & Honors

• **Reliance Foundation Undergraduate Scholar** (2023) – Qualified RF-UG Aptitude test, top 5000 awardees nationally.
• **Best Young Scientist Speaker on Nanotechnology** (2019) – World Science Conference, Jadavpur University.

Professional Experience

Freelance Web Developer & Designer *2024–2025*
Designed and developed the website for ChemActiva Innovations Pvt. Ltd. (nanocellulose startup, DST-NIDHI PRAYAS & HDFC Parivartan grantee). Also led web development for Anicon 3.0 (Inquivesta XI) and the EFAML Lab page at IISER-K.

Private Educator & Technical Trainer *2018–Present*
Self-employed, Kolkata. Courses in CS, Physics, Chemistry for ICSE/CBSE/WB Board. Mentored 50+ students for competitive exams.

Technical Consultant *2021–Present*
Self-employed, Kolkata. HPC solutions, system optimization, BIOS/UEFI config. 50+ custom builds with 100% client satisfaction.

Published Author – *MindScapes* (ISBN: 978-9389923209) *2020*
Published creative anthology. Conducted workshops on technical and creative writing.

Education

Indian Institute of Science Education and Research, Kolkata
2022–2027 (expected)
BS-MS (Chemistry Major, Computer Science Minor) CGPA: 8.2

Calcutta University *2021–2022*
B.Sc Honours in Physics (1st Year Only) CGPA: 8.308

Jodhpur Park Boys' High School (WBCHSE) *2019–2021*
Higher Secondary – Physics, Mathematics, Chemistry, Computer Science 83%

The New Horizon High School (WBBSE) *2009–2019*
Secondary Level Schooling (English Medium) 83.75%

Technical Skills

Core Competencies: Technical Leadership, Research & Development, Algorithm Development, Strategic Planning, Event Management, Mentoring, Public Speaking
Research Computing: Information Retrieval (Apache Lucene, BM25), Bio-Informatics, Molecular Dynamics, DFT Computations, Reciprocal Rank Fusion, Information Theory
Languages: Python, C/C++, Java, Rust, Fortran, QBASIC, GWBASIC
Python – Core & Scientific: Numpy, Pandas, SciPy, Matplotlib, Plotly, Scikit-learn, OpenCV, PyTorch, TensorFlow
Python – NLP & ML: HuggingFace Transformers, LangChain, NLTK, SpaCy, FAISS, ChromaDB, Selenium, BeautifulSoup
Python – Document Processing: PyMuPDF, EasyOCR, Streamlit, Manim, fake_useragent
LLMs & Embeddings: Qwen2/3-4B, Gemma (1B/27B), DeepSeek, LLaMA-3.2, Nomic Embeddings, Mistral-OCR, Indic-Conformer
RAG/IR: Dense+Sparse Retrieval, BM25, RRF, Vector DBs (FAISS, ChromaDB), Hybrid Search
Scientific Software: LAMMPS, VMD, Gaussian, Origin Pro, Scilab, DFT
Bio-Informatics: PyMol, ChimeraX, PyDock, AutoDock Vina
Digital Pathology: HistoQC, PathProfiler, GrandQC, HistoROI, FASTPathology
Web & Dev Stack: Django, Flask, React, Three.js, HTML/CSS/JS, MongoDB, MySQL, SQLite3, Git, Docker, CMake, Make
GUI Development: GTK4 (C), Qt (C++), Glade
OS & Server: Linux, SSH, OpenSSL, CUDA (Python), Server Handling
Lab Instrumentation: UV-Vis, ATR-FTIR, TGA, DSC, XRD, GC-MS, SEM, TEM, AFM, Fluorimeter, Optical Bench, Column Chromatography
Tools: LaTeX, Bash, Figma, UI/UX Design, HTML E-mailing

Invited Talks & Jury Positions

Technical Talk – Running & Optimizing Local LLMs *Aug 2025*
Slashdot Student Chapter, IISER-K. Hands-on session: hardware setups, quantization methods, embedding pipelines, vector databases, and optimization techniques.

Invited Talk – Sustainability, AI & Emerging Technologies *Aug 2025*
Valence – Chemistry Society, IISER-K. Interdisciplinary talk bridging sustainability science with AI-driven modelling, retrieval systems, and computational chemistry.

Domain Judge – Computer Science Events *July 2025*
La Martiniere for Boys Annual Science Fest, Kolkata. External judge for project demonstrations, live coding, and problem-solving rounds.

Leadership & Community Impact

• **Office Bearer**, Slashdot – Programming & Design Club, IISER-K (Aug 2025–). Leading workshops, hackathons, and technical sessions to foster coding culture.
• **Office Bearer**, Valence – Chemistry Club, IISER-K (Aug 2025–). Organizing seminars, student talks, and departmental events.
• **Organizer**, Qiskit Fallfest 2025 (IBM Quantum sponsored, one of 150 selected institutes globally). Hands-on workshops on Qiskit SDK and quantum computing.
• **Event Lead**, Anicon 3.0 & 2.0 (500+ participants); Supra-Molecular Discussions 2024; GIAN Courses on Soft-Oxometalates and X-Ray Crystallography.
• **Social Impact**: COVID-19 relief coordinator (2020–21); Ek-Pehal education mentoring initiative.
• District-level Debate/Quiz finalist | Shotokan Karate | 4th Year Art & Painting Student (Stroke Art & Portraits).

References available upon request